

NLP_Practical_05

April 7, 2026

0.0.1 Step-1: Import Libray

```
[1]: import pandas as pd
```

0.0.2 Step-2: Data

```
[2]: data = [  
    {"word": "playing", "root": "play"},  
    {"word": "walked", "root": "walk"},  
    {"word": "happiness", "root": "happy"},  
    {"word": "unlucky", "root": "lucky"}  
]  
  
df = pd.DataFrame(data)  
df
```

```
[2]:      word  root  
0   playing  play  
1    walked  walk  
2 happiness  happy  
3   unlucky  lucky
```

0.0.3 Step-3: Function to Analyze Morpholog

```
[3]: def analyze_morphology(word, root):  
  
    # Case 1: Suffix added (playing = play + ing)  
    if word.startswith(root):  
        delete = ""  
        add = word[len(root):]  
  
    # Case 2: Prefix added (unlucky = un + lucky)  
    elif word.endswith(root):  
        delete = ""  
        add = word[:-len(root)]  
  
    # Case 3: Word modified (happiness = happy + happi + ness)  
    else:
```

```

    i = 0
    while i < min(len(word), len(root)) and word[i] == root[i]:
        i += 1

    delete = root[i:]
    add = word[i:]

    return delete, add

```

0.0.4 Step-4: Create Add-Delete Table

```

[4]: rows = []

for item in data:
    word = item["word"]
    root = item["root"]

    delete, add = analyze_morphology(word, root)

    rows.append({
        "Word": word,
        "Root": root,
        "Delete": delete,
        "Add": add
    })

```

```

[5]: df = pd.DataFrame(rows)
print("Add-Delete Table:\n")
print(df)

```

Add-Delete Table:

	Word	Root	Delete	Add
0	playing	play		ing
1	walked	walk		ed
2	happiness	happy	y	iness
3	unlucky	lucky		un

```
[ ]:
```